



UNIVERSITY OF SAN-JOSE RECOLETOS

Engineering Mathematical Society

A.Y. 2026 - 2027



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Basic Derivatives

Practice Problems

1. Find the Derivative of $f(x) = x^5 - 4x^3 + 2x - 7$
2. Find $\frac{dy}{dx}$ if $y = \frac{3}{x^2}$
3. Differentiate $y = 4\sqrt{x}$
4. Find the Derivative of $f(x) = (x^3 + 1)(e^x)$
5. Find the Derivative of $f(x) = 3x^3 - \frac{2}{x^2} + \sqrt[3]{x}$
6. Find $\frac{dy}{dx}$ if $y = x^3(5x^2 - 2)$
7. Differentiate using the Quotient Rule: $y = \frac{x+5}{x-2}$
8. Differentiate $y = \frac{2x^2}{3x+1}$
9. Find the Derivative of $f(x) = (x^3 - 2x + 1)^4$
10. Find the Derivative of $f(x) = (4x^2 - 3)^6$ using chain rule.



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Basic Derivatives
Step-by-step Solution

1. Find the Derivative of $f(x) = x^5 - 4x^3 + 2x - 7$

- **Rule:** Use the Power Rule ($\frac{d}{dx}[x^n] = nx^{n-1}$) on each term. The derivative of a constant is 0.
- Step 1: Differentiate $x^5 = 5x^{5-1} = 5x^4$
- Step 2: Differentiate $-4x^3$ by multiplying the exponent by the coefficient:
 $-4 \times 3x^{3-1} = -12x^2$
- Step 3: Differentiate $2x: 2x^{1-1} = 2x^0 = 2$
- Step 4: Differentiate $-7: \frac{d}{dx}[-7] = 0$

Answer: $f'(x) = 5x^4 - 12x^2 + 2$

2. Find $\frac{dy}{dx}$ if $y = \frac{3}{x^2}$

- **Rule:** Rewrite the expression with a negative exponent, then apply the Power Rule.
- Step 1: Rewrite the equation using negative exponent rule: $\frac{3}{x^2} = 3x^{-2}$
- Step 2: Apply the Power Rule: $3 \times -2x^{-2-1} = -6x^{-3}$
- Step 3: Rewrite the answer with a positive exponent using reciprocal exponent rule for the final answer: $-6x^{-3} = -\frac{6}{x^3}$



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Answer: $y' = -\frac{6}{x^3}$

3. Differentiate $y = 4\sqrt{x}$

- **Rule:** Rewrite the radical as a fractional exponent, then apply the Power Rule.
- Step 1: Rewrite the equation: $y = 4\sqrt{x} \Rightarrow y = 4x^{\frac{1}{2}}$
- Step 2: Apply the power rule: $4 \times \frac{1}{2}x^{\frac{1}{2}-1} = 2x^{-\frac{1}{2}}$
- Step 3: Move the negative exponent to the denominator using the reciprocal exponent

rule and rewrite as a radical: $2x^{-\frac{1}{2}} = \frac{2}{x^{\frac{1}{2}}} = \frac{2}{\sqrt{x}}$

Answer: $y' = \frac{2}{\sqrt{x}}$

(**Note:** In this item, you should zoom IN the steps for clarity and to avoid the wrong process.)

4. Find the Derivative of $f(x) = (x^3 + 1)(e^x)$

- **Rule:** Use the Product Rule: $\frac{d}{dx} [uv] = (u)(dv) + (v)(du)$. Let $u = x^3 + 1$ and $v = e^x$
- Step 1: Find the derivatives of u and v.

$$du = 3x^{3-1} + 0 = 3x^2 \text{ and } dv = e^x \text{ (the derivative of } e^x \text{ is still } e^x \text{).}$$

- Step 2: Plug into the Product Rule formula:

$$f'(x) = (x^3 + 1)(e^x) + (e^x)(3x^2) = (x^3 e^x + e^x) + (3x^2 e^x) = (x^3 e^x + 3x^2 e^x + e^x)$$



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- Step 3: Factor out the common e^x to simplify: $f'(x) = e^x(x^3 + 3x^2 + 1)$

Answer: $f'(x) = e^x(x^3 + 3x^2 + 1)$

5. Find the Derivative of $f(x) = 3x^3 - \frac{2}{x^2} + \sqrt[3]{x}$

- **Rule:** Rewrite all terms with exponents in the numerator, then use the Power Rule .

- Step 1: Rewrite the equation: $f(x) = 3x^3 - \frac{2}{x^2} + \sqrt[3]{x} \Rightarrow f(x) = 3x^3 - 2x^{-2} + x^{\frac{1}{3}}$

- Step 2: Differentiate each term:

$$\frac{d}{dx} [3x^3] = 3 \times 3x^{3-1} = 9x^2$$

$$\frac{d}{dx} [-2x^{-2}] = -2 \times -2x^{-2-1} = 4x^{-3}$$

$$\frac{d}{dx} [x^{\frac{1}{3}}] = \frac{1}{3}x^{\frac{1}{3}-1} = \frac{1}{3}x^{-\frac{2}{3}}$$

$$\text{So far, } f'(x) = 9x^2 + 4x^{-3} + \frac{1}{3}x^{-\frac{2}{3}}$$

- Step 3: Rewrite the equation using reciprocal exponent rule:

$$4x^{-3} = \frac{4}{x^3}$$

$$\frac{1}{3}x^{-\frac{2}{3}} = \frac{1}{3x^{\frac{2}{3}}}$$

$$f'(x) = 9x^2 + \frac{4}{x^3} + \frac{1}{3x^{\frac{2}{3}}}$$

Answer: $f'(x) = 9x^2 + \frac{4}{x^3} + \frac{1}{3x^{\frac{2}{3}}}$



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6. Find $\frac{dy}{dx}$ if $y = x^3(5x^2 - 2)$

- Rule: Expand the expression first to avoid the Product Rule, then use the Power Rule.
- Step 1: Distribute x^3 into the parentheses to get $y = 5x^5 - 2x^3$
- Step 2: Apply the Power Rule to $5x^5$: $\frac{d}{dx} [5x^5] = 5 \times 5x^{5-1} = 25x^4$
- Step 3: Apply the Power Rule to $-2x^3$: $\frac{d}{dx} [-2x^3] = -2 \times 3x^{3-1} = -6x^2$

Answer: $y' = 25x^4 - 6x^2$

7. Differentiate using the Quotient Rule: $y = \frac{x+5}{x-2}$

- **Rule:** The Quotient Rule states that $\frac{d}{dx} \left[\frac{u}{v} \right] = \frac{vdu - u dv}{v^2}$, let $u = x + 5$ and $v = x - 2$
- Step 1: Find du and dv :

$$u' = \frac{d}{dx} [x + 5] = 1$$

$$v' = \frac{d}{dx} [x - 2] = 1$$

- Step 2: Substitute to the formula and solve:

$$y' = \frac{(x-2)(1) - (x+5)(1)}{(x-2)^2}$$

$$y' = \frac{x-2 - (x+5)}{(x-2)^2}$$

$$y' = \frac{x-2 - x-5}{x^2 - 4x + 4}$$



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$$y' = -\frac{7}{x^2-4x+4}$$

Answer: $y' = -\frac{7}{x^2-4x+4}$

8. Differentiate $y = \frac{2x^2}{3x+1}$

- **Rule:** Use the Quotient Rule. Let $u = 2x^2$ and $v = 3x + 1$.

- Step 1: Find u' and v' .

$$u' = \frac{d}{dx} [2x^2] = 4x$$

$$v' = \frac{d}{dx} [3x + 1] = 3$$

- Step 2: Substitute into the formula and solve:

$$y' = \frac{(3x+1)(4x) - (2x^2)(3)}{(3x+1)^2}$$

$$y' = \frac{(12x^2+4x) - (6x^2)}{(3x+1)^2}$$

$$y' = \frac{12x^2+4x-6x^2}{9x^2+6x+1}$$

$$y' = \frac{6x^2+4x}{9x^2+6x+1}$$

Answer: $y' = \frac{6x^2+4x}{9x^2+6x+1}$

9. Find the Derivative of $f(x) = (x^3 - 2x + 1)^4$



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- **Rule:** Use the Chain Rule $\frac{d}{dx}[g(h(x))] = g'(h(x)) \times h'(x)$, taking the derivative of the "outside" function and multiplying it by the derivative of the "inside" function.
- Step 1: Differentiate the outside function ($g(x)$) while leaving the inside function ($h(x)$) unchanged:

$$\Rightarrow \text{Let } g(x) = (h(x))^4$$

$$\Rightarrow \text{Let } h(x) = x^3 - 2x + 1$$

$$\Rightarrow g(h(x)) = (x^3 - 2x + 1)^4$$

Differentiate the outside function:

$$g'(h(x)) = 4(x^3 - 2x + 1)^3$$

- Step 2: Differentiate the inside function ($h(x)$):

$$h'(x) = 3x^2 - 2$$

- Step 3: Multiply Step 1 & Step 2 and solve.

$$\frac{d}{dx}[g(h(x))] = 4(x^3 - 2x + 1)^3 \times (3x^2 - 2)$$

$$\frac{d}{dx}[g(h(x))] = (x^3 - 2x + 1)^3(12x^2 - 8)$$

$$\text{Answer: } f'(x) = (x^3 - 2x + 1)^3(12x^2 - 8)$$

10. Find the Derivative of $f(x) = (4x^2 - 3)^6$ using chain rule.



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- Rule: Use the chain rule, establish the outside function and the inside function.

Outside function $\Rightarrow g(h(x)) = (h(x))^6$

Inside function $\Rightarrow h(x) = (4x^2 - 3)$

- Step 1: Differentiate the outside function:

$$g'(h(x)) = 6(h(x))^5 = 6(4x^2 - 3)^5$$

- Step 2: Differentiate the inside function:

$$h'(x) = 8x$$

- Step 3: Multiply step 1 & 2 and solve.

$$f'(x) = 6(4x^2 - 3)^5 \times (8x)$$

$$f'(x) = (4x^2 - 3)^5 (48x)$$

Answer: $f'(x) = (48x)(4x^2 - 3)^5$